

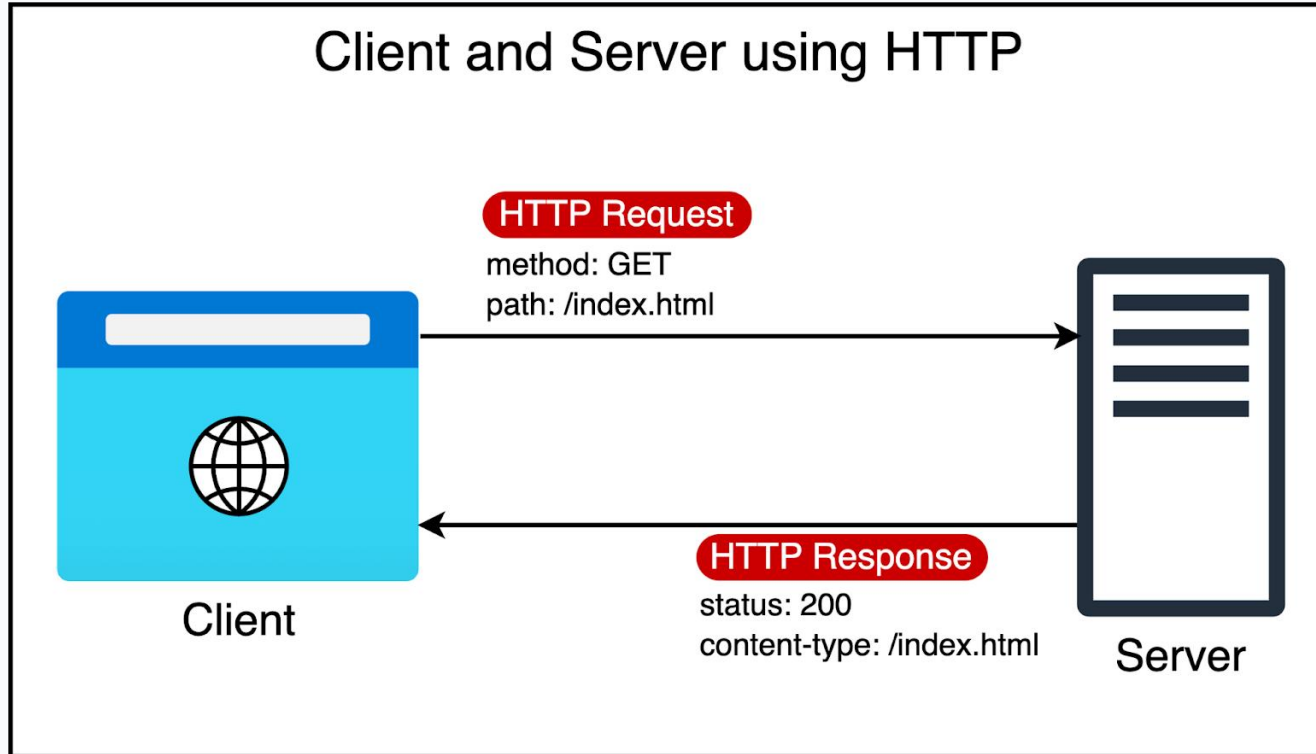
TRAINING MODULE

client-server architecture

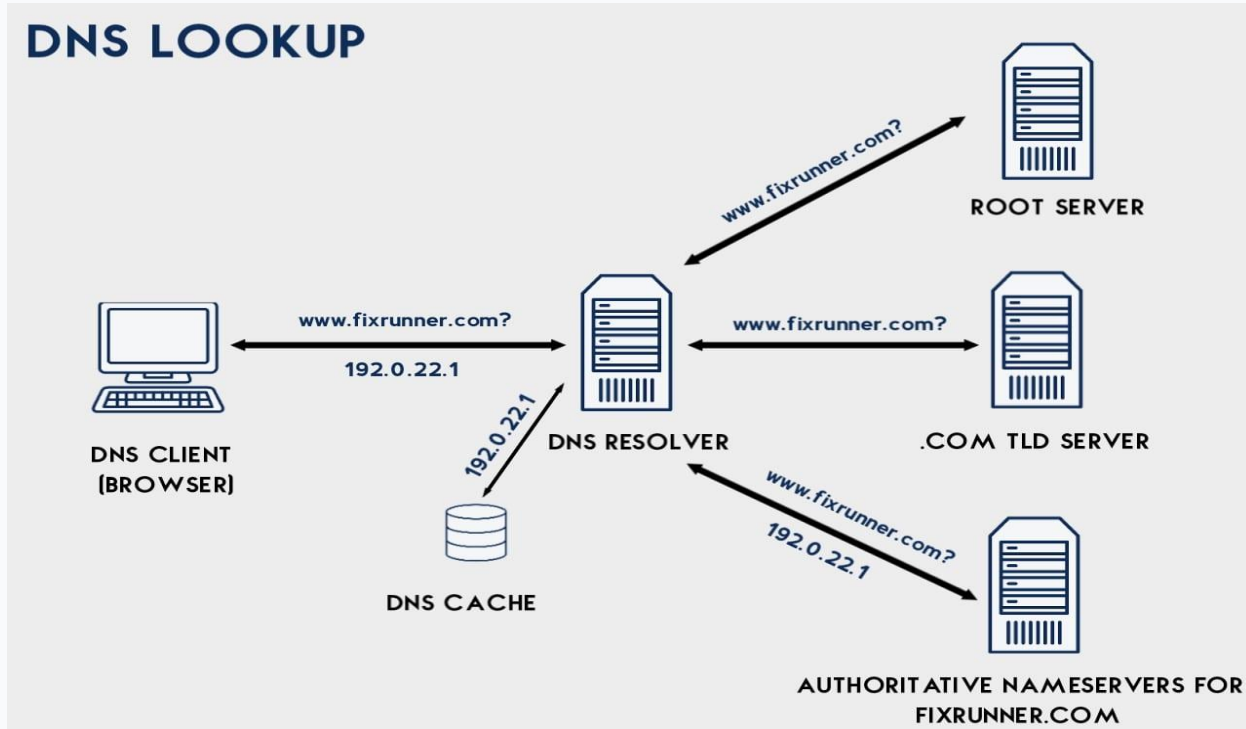
explore start of connection between client & server

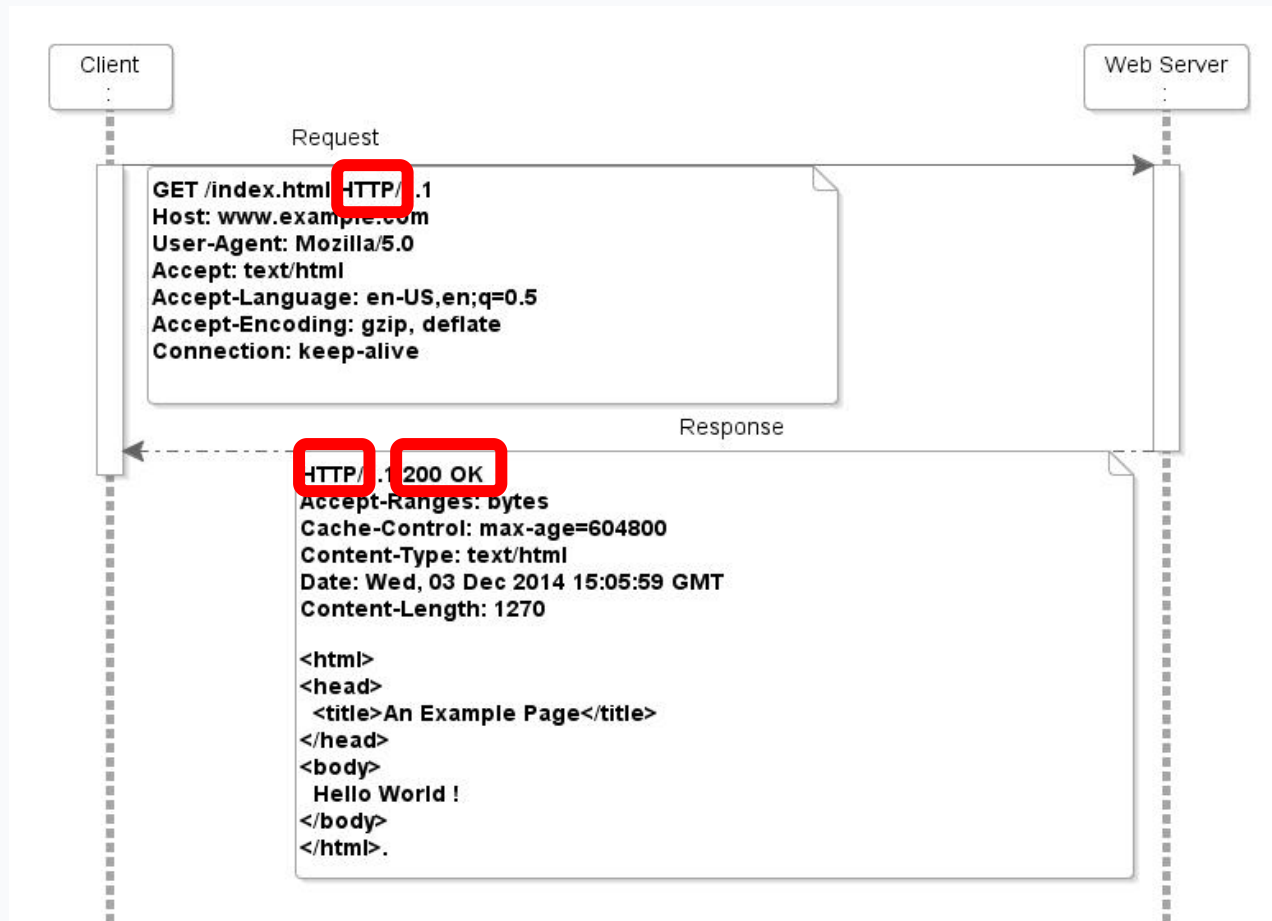


client-server architecture

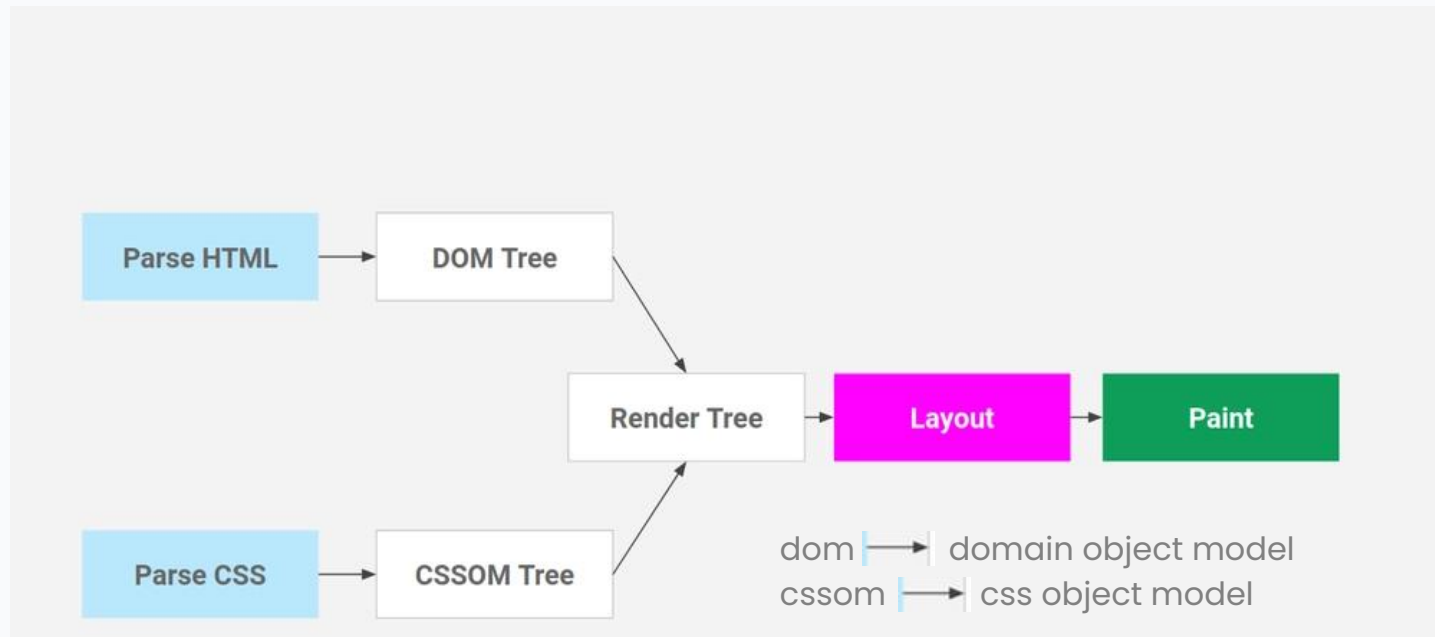


DNS resolving





Page redndering (browser effect)



TRAINING MODULE

HTML Basics

Building the base of web page



DOCUMENT ANATOMY

- ✓ **DOCTYPE** declaration informs the browser about the HTML version.
- ✓ **<html>** is the root element wrapping all content.
- ✓ **<head>** contains metadata, titles, and resource links.
- ✓ **<meta charset="UTF-8"> search !!**
- ✓ **<body>** holds the visible user interface elements.

invalid HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title> Page Title </title>
  </head>
  <body>
    <h1> Hello World </h1>
  </body>
</html>
```

| Common Text Tags – headings

<h1> Hello, world</h1>

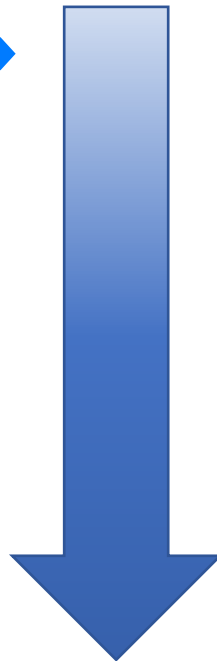
<h2> Hello, world</h2>

<h3> Hello, world</h3>

<h4> Hello, world</h4>

<h5> Hello, world</h5>

<h6> Hello, world</h6>



Font size

| Common Text Tags – Text Content

`<p>`

A "Hello World" sentence is the classic phrase used in computing to test if a system is working properly.

`</p>`

`<span> Hello, world</span>`

| Common Text Tags – Lists

1) UNORDERED LIST

`<ul>`

`<li> item 1 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`</ul>`

| Common Text Tags – Lists

2) ORDERED LIST

`<ol>`

`<li> item 1 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`<li> item 2 </li>`

`</ol>`

| Common Button Tags – Clickable Buttons

a) normal button (we'll customize its function in later in js)

```
<button> click me </button>
```

b) anchor (redirect to url)

```
<a href="https://www.youtube.com/"> click me </a>
```

| Common Media Tags – Image

```

```

note that `img` is “self closing tag” or “empty tag”

`<img> </img>` ❌

`<img />` ✅

| Common Media Tags – Video

This is your Task 🔥🔥

| Form Tag

```
<form method="GET"> <!-- OR method="POST" -->  
<label for="name"> Name :</label>  
<input type="text" id="name" placeholder="please enter your name"/>  
<input type="submit" value = "submit"/>  
</form>
```

TAGS & ATTRIBUTES



Container Tags

Elements that wrap content with both opening and closing tags. They define a clear scope.

```
<p>Hello</p>
```



Empty Tags

Self-closing elements that don't hold text content. Used for structural or media assets.

```

```



Attributes

Key-value pairs within tags that provide extra configuration or metadata.

```
href="https://..."
```

TRANSITIONING TO SEMANTIC HTML

Moving beyond generic containers to elements that describe their **purpose** and **meaning** to both browsers and developers.

? Non-Semantic

<div>

Tells us nothing about the content inside. Just a generic box.

✓ Semantic

<header>

<article>

<footer>

Clearly defines the role of the content. Improves structure and clarity.

WHY SEMANTIC HTML?



Accessibility

Enables **screen readers** and assistive technologies to navigate content logically for users with visual impairments.



SEO

Helps **search engines** understand the priority and context of your content, improving site ranking and discoverability.



Clarity

Creates **maintainable code** that is easier for developers to read, debug, and scale across large enterprise projects.

ESSENTIAL LAYOUT ELEMENTS

<header>

Introductory content, logos, or site-wide navigation wrappers.

<nav>

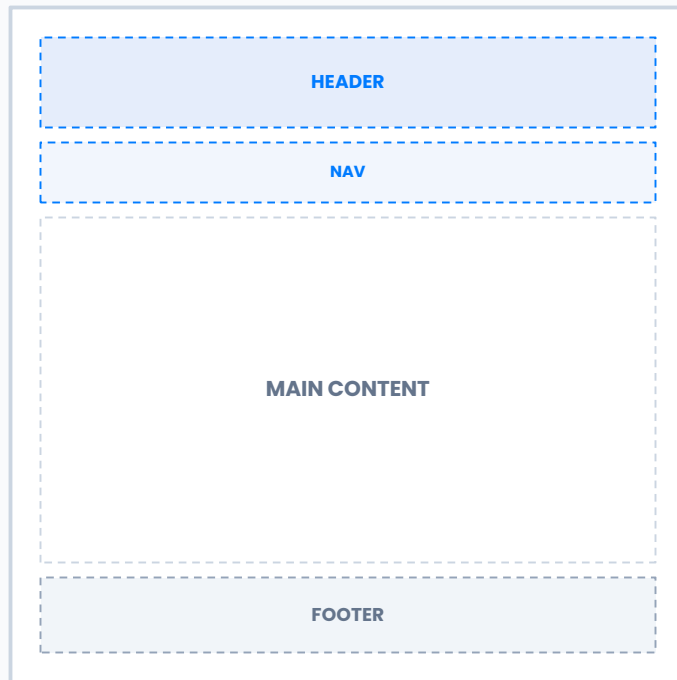
A section intended for primary navigation links.

<main>

The dominant, unique content of the document body.

<footer>

Closing information, copyrights, and contact details.



STRUCTURING CONTENT



<section>

Groups **related content** together within a document, typically with its own heading.



<aside>

Contains content **indirectly related** to the main topic, such as sidebars or callouts.



<article>

Represents a **self-contained**, independent piece of content like a blog post or news story.



<figure>

Used for **media content** (images, diagrams) along with an optional <figcaption>.

KEY TAKEAWAYS

Structure vs. Meaning

HTML provides the **foundation**, but Semantic HTML provides the **context** for browsers and users.

Accessibility First

Always prioritize **semantic tags** to ensure your web applications are inclusive and SEO-friendly.

Continuous Practice

Refactor existing projects to replace generic **div containers** with meaningful structural elements.

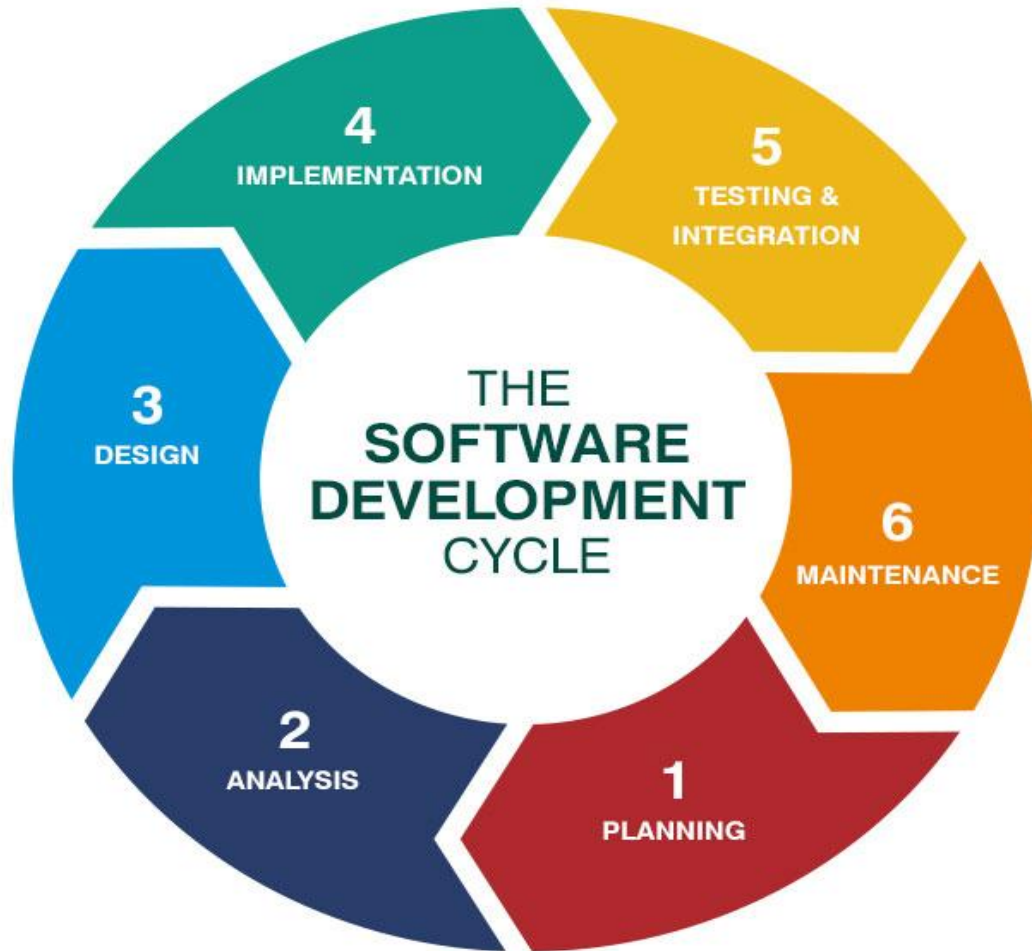
Q&A

Thank you for
attending

Software Development Life Cycle (SDLC)

The Blueprint for Software Success





FINAL MISSION: MEET THE DEVELOPER

Apply everything you've learned to build a professional **Personal Introduction** page about you that showcases your skills and personality.

Structural Integrity

Include a full `<!DOCTYPE html>`, `<head>`, and `<body>`.
(use meta & video & img tags)

Semantic Hierarchy

Organize content using `<header>`, `<nav>`, `<main>`, and `<footer>`.

The Skills Table

Create a **complex table** for your technical skills

My card info		
my skills	HTML	CSS
	JS	React
My name	Abdelrahman Ali	

Table example

SUCCESS CRITERIA

- ✓ Proper tag nesting
- ✓ Semantic tag usage
- ✓ Complex table logic
- ✓ Clean, readable code

